

EXAMINATION QUESTION PAPER
Question 2 of 4

Topic: Convolution in Digital Signal Processing

Total Marks: 25

Instructions: Answer ALL parts. Show all working clearly. Marks are allocated as shown beside each part.

QUESTION 2

(a) Convolution: Definition and Fundamental Concepts

(i) Define convolution in the context of discrete-time digital signal processing. Write the notation used to represent the convolution of an input signal $x[n]$ with an impulse response $h[n]$, and identify the three fundamental signals related by convolution in a linear system.

[4 marks]

(ii) Define the **delta function** $\delta[n]$ (unit impulse) mathematically. State its key property that makes it essential to DSP, and give a worked example of how a sample with value -3 at index 8 can be expressed using the delta function.

[3 marks]

(iii) Explain what the **impulse response** $h[n]$ of a linear system is. Why does knowing the impulse response mean you know *everything* about the system? List two alternative names for the impulse response used in different application areas.

[3 marks]

(b) Algorithms for Computing Convolution

(i) Describe the **Input Side Algorithm** for convolution. In your answer, outline the four-step process and explain what the final output represents in terms of scaled and shifted impulse responses.

[3 marks]

(ii) The **Output Side Algorithm** (convolution sum) is defined as:

$$y[n] = \sum h[j] \cdot x[n - j], j = 0 \text{ to } M-1$$

Identify each variable in this formula. Explain the significance of the index $n - j$ and the term 'flipping' in the context of the output side algorithm.

[3 marks]

(iii) An input signal $x[n]$ has **N = 120** samples and an impulse response $h[n]$ has **M = 45** samples. Calculate the length of the output signal $y[n]$, stating the formula used. Also calculate the total number of multiply-accumulate operations required for direct convolution.

[3 marks]

(c) Common Impulse Responses and Mathematical Properties

(i) Complete the table below by providing the impulse response $h[n]$ and the resulting output $y[n]$ for each system behaviour listed.

System Behaviour	Impulse Response $h[n]$	Output $y[n]$
Identity (no change)		
Amplify by factor k		
Delay by s samples		
Echo (delay s , scale k)		
First Difference (discrete derivative)		

[5 marks]

(d) Correlation and Computational Speed

(i) Distinguish between **convolution** and **correlation**. Explain how correlation can be expressed using convolution mathematics, and state the mathematical relationship between the two operations.

[3 marks]

(ii) Briefly describe **one** real-world application of cross-correlation and **one** application of autocorrelation, explaining how each uses the operation to solve a practical signal processing problem.

[3 marks]

Marks Allocation Summary

Part	Sub-questions	Marks
(a)	(i) 3 + (ii) 2 + (iii) 2	7
(b)	(i) 3 + (ii) 3 + (iii) 3	9
(c)	(i) 5	5
(d)	(i) 2 + (ii) 2	4
TOTAL		25

— End of Question 2 —